🚀 CORE JAVA ROADMAP (0 → Strong Level)

🟢 PHASE 1: Java Basics (Foundation)

⏳ Time: 2–3 Weeks

✅ 1. Introduction to Java

- What is Java?
- JDK, JRE, JVM
- How Java works
- Install JDK
- Write first program
- Compilation process

✅ 2. Basic Syntax

- Variables & Data Types
- Operators
- Type casting
- Input (Scanner class)
- Comments

✅ 3. Control Statements

- if, if-else
- switch
- for loop
- while loop
- do-while
- break & continue

🎯 Practice:
Make small programs like:

- Prime number
- Palindrome
- Pattern printing
- Calculator

🟡 PHASE 2: OOP Concepts (Very Important 🔥)

⏳ Time: 3–4 Weeks

This is the HEART of Java.

✅ 4. Classes & Objects

- Creating class
- Object creation
- Constructors
- Methods

✅ 5. OOP Principles

- Encapsulation
- Abstraction
- Inheritance
- Polymorphism

🎯 Interview favorite topic ⚠

✅ 6. this & super keyword

✅ 7. Method Overloading & Overriding

✅ 8. Access Modifiers

- public
- private
- protected
- default

🟠 PHASE 3: Important Core Concepts

⏳ Time: 3 Weeks

✅ 9. Strings (Very Important)

- String class
- StringBuilder
- StringBuffer
- Immutable concept

✅ 10. Arrays

- 1D array
- 2D array
- Array operations

✅ 11. Wrapper Classes

- Autoboxing
- Unboxing

✅ 12. Exception Handling

- try-catch
- finally
- throw & throws
- Custom exception

🔥 Interview important

✅ 13. Collections Framework (Very Important 🚀)

- List (ArrayList, LinkedList)
- Set (HashSet)
- Map (HashMap)
- Iterator
- Comparable vs Comparator

⚠ Most asked in interviews

🔵 PHASE 4: Advanced Core Java Concepts

⏳ Time: 2–3 Weeks

✅ 14. Multithreading

- Thread class
- Runnable interface
- Synchronization
- Thread lifecycle

✅ 15. File Handling

- File class
- FileReader
- FileWriter
- BufferedReader

✅ 16. Java 8 Features (Very Important 🔥)

- Lambda expressions
- Functional interface
- Stream API
- Optional class

Modern companies expect this.

🔴 PHASE 5: Practice & Projects

⏳ Time: Ongoing

Build small console projects:

1️⃣ Bank Management System
2️⃣ Student Management System
3️⃣ Library System
4️⃣ ATM Machine Simulation

Upload on GitHub.

📚 Daily Study Plan (If You Are in 2nd Year)

- 1.5 – 2 hours daily
- 5 days coding practice
- 2 days revision

Finish in 2–3 months easily.

🎯 Interview Preparation

After finishing:

✓ Practice 100+ Java interview questions
✓ Solve basic DSA problems
✓ Understand memory management
✓ Understand JVM architecture